# JAMES ALEXANDER

## SENIOR DATA SCIENTIST – MCLAREN RACING

Applied ML Engineer working across the whole stack: computer vision, anomaly detection, agentic AI, and data engineering.
Years of experience across defense, finance and manufacturing, with one granted patent (GB2631290) and a second pending.


## CONTACT DETAILS

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## SKILLS

- COMPUTER VISION - Expert in computer vision (OpenCV, Ultralytics, PyTorch). Granted a patent for automated construction-schematic cost estimation. Second patent pending for a **zero-shot, CPU-only anomaly-detection framework**: training-free, real-time at **>1000 FPS**, with accuracy **competitive with trained-GPU SOTA** on public industrial-video benchmarks.

- DATA SCIENCE AND MACHINE LEARNING - During my time as a data scientist I have applied a variety of techniques, such as regression, clustering, dimensionality reduction and predictive time-series modelling. I am comfortable with **PyTorch, Numpy, Pandas, Scikit-Learn and Scipy** data processing modules and have delivered visuals to clients using **Matplotlib, Plotly and Seaborn visualization libraries**

- PROGRAMMING - Fluent in **Python** and **LaTeX.** Competent **with C, C#, SQL and R** at an intermediate level and **React** at a basic level. Experience with both Windows and Linux, as well as **Docker** environments. Competent with **Git.**

- CLOUD COMPUTING - Experience with **AWS**. Thorough knowledge of Connect, Kinesis, Sagemaker, Bedrock, Lambda and Lex. Built a live call analytics platform capable of monitoring calls and providing AI assistance using a knowledge base to assist in policy fetching, suggested responses, sentiment analysis and category alerts.

- GEN AI AND AGENTIC AI - I have built and deployed **RAG Pipelines** within AWS from scratch, using **Bedrock**, with near zero latency, in addition to building a customized **Agentic Framework** tailored to small, local models.

## EXPERIENCE

**SENIOR DATA SCIENTIST | CONTRACTOR –** McLaren Racing Via PurpleSector Subcontracting
August 2025 – Current
As a senior developer in PurpleSector, the subcontractor for the McLaren Accelerator program, I am part of the branch that consults for our sponsors. I have established myself as a leading expert in computer vision and data analysis and am now the sole developer responsible for computer vision support across all client work.
In my time here, I have:

- Ensured repeat business a global food manufacturer as the lead developer on the initial starting project. This was achieved by setting up hardware and within 3 days developing an anomaly detection suite capable of operating on lines running at 1000+ parts per minute. These anomalies were then analyzed to **reduce unexpected downtime by 5x**. **We achieved an ROI of approximately 10x, saving the customer well over $10m per year.**
- Ensured repeat business, and then led the subsequent project, for one of the world's largest spirit companies. Similarly, we arrived on site and within 24 hours were able to demonstrate our ability to develop and deploy a computer vision system capable of identifying and quantifying deformity in bottle caps. During this project we have **improved eOEE by ~20%** for they're worst performing product.

- Led IP development and product development. I created a zero-shot generalized anomaly detection pipeline which currently is being patented and have further expanded that to zero-shot generalized anomaly classification. As a result I am now the sole Data Scientist supporting across all projects, ensuring anomaly detection runs on the edge, with CPU only, at frame rates of 150+ in 1080p.

**MACHINE LEARNING ENGINEER -** Intrum/Capquest
January 2024 – August 2025
This role focused on optimizing call centers using AWS. Using Amazon Connect all calls were processed, transcribed and analyzed in the cloud. Using insights from this data I have:

- Created a quality assurance dashboard capable of giving a broad company-wide view on the performance of agents and customer satisfaction, highlighting key areas for future training.
- Optimized agent time by creating an auto-summarizer based on the call transcripts using Gen AI.
- Built an AI assistant chatbot that can also provide real time agent assistance while on call, capable of retrieving policies, reminding agent of key requirements and suggesting responses based on the customer's last sentence.

**SENIOR COMPUTER VISION ENGINEER -** Ensign Advanced Systems Ltd
September 2022 – January 2024

- Patent granted (GB2631290) for smart auto-counting methodology described as "revolutionary" by customers.
- Gained experience in a managerial role as project lead, managing both developers responsible for the front-end GUI and back-end infrastructure.

**DATA SYSTEMS ANALYST -** Chetwood Financial Ltd
September 2021 – September 2022

- Provided support to the credit card business via data analytics and data warehousing.
- Created a repayment calculator for credit cards and saw it through to deployment on iOS/Android.
- Restructured the notification system for the credit card application.
- Created reports for investors using MySQL, PostgreSQL and python using infrastructure such as AWS Redshift, RDS and SageMaker.

**DEVELOPMENT ENGINEER -** Excelitas/Qioptiq
June 2019 – September 2021

- Developed a range of skills relevant to opto-electronics, embedded systems and digital video processing.
- Acted as technical lead for the Video Processing Roadmap; developing a series of post-processing solutions, using sci-kit learn and OpenCV, and seeing through both the software and hardware deployment.
- Acted as technical lead for the Artificial Intelligence Roadmap; focused on research into deep-learning (using TensorFlow/Keras) and how it is of benefit in the industry, with the aim to create, train and deploy and embedded AI solutions.

**DATA SCIENTIST/SYSTEMS ENGINEER -** BAE Systems Maritime Services
September 2017 – June 2019

- Data Scientist - Developed foundational knowledge in the field of Data Science. I produced an algorithm which allowed predictions of correlations between two events over a given time period – referred to as market basket analysis.
- Systems Engineer - Integrating theoretical knowledge into the maritime services industry via a Year in Industry. I created a modelling solution for naval vessel power consumption. The advanced level of precision had not been seen in the industry before.

## QUALIFICATIONS

**BSC PHYSICS DEGREE** – The University of Exeter – 2015 – 2019 Incl Year in Industry

**A-LEVELS** –Ysgol Eirias – 2013 – 2015
Mathematics (A*), Further Mathematics (A), Physics (B)

**FULL UK DRIVING LICENCE**

References available upon request.